

IBRAHIM ALI GHANIM

MECHATRONICS ENGINEER | ARTIFICIAL INTELLIGENCE, COMPUTER VISION & SOFTWARE DEVELOPMENT

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PROFESSIONAL SUMMARY

Mechatronics Engineering graduate (2026), ranked 9th out of 90 students (Top 10%), with applied experience in artificial intelligence, computer vision, real-time software, APIs, embedded systems, and industrial AI integration. Built projects using Python, YOLOv8, OpenCV, Scikit-learn, FastAPI, Flask, Node-RED, and ESP32 for visual inspection, anomaly detection, intelligent control, and human-machine interaction.

EDUCATION

B.Sc. in Mechatronics Engineering | University of Baghdad - Al-Khwarizmi College of Engineering | June 2026

Ranked 9th out of 90 Mechatronics Engineering students (Top 10% of the cohort)

TECHNICAL SKILLS

AI, Machine Learning & Computer Vision: Python, Scikit-learn, YOLOv8, OpenCV, MediaPipe, Isolation Forest, MATLAB Vision, Object Detection, Anomaly Detection, Temporal Analysis, ROI/Edge/Orientation Analysis, Prompt Engineering

Software & API Development: JavaScript, HTML/CSS, FastAPI, Flask, REST APIs, JSON/HTTP, Telethon, Real-Time Dashboards, Data Processing

Industrial AI & Integration: Node-RED, Factory I/O, Modbus TCP, Ignition 8.3, OPC UA, SCADA/HMI Integration, Alarm & Event Data

Embedded Systems & Robotics: ESP32, Arduino, Mobile Robotics, Encoder Odometry, Ultrasonic Obstacle Avoidance, FSM Navigation, Wi-Fi Telemetry, Camera & Sensor Integration

Interactive 3D & Web Technologies: Three.js, WebGL, Rapier Physics, GLB/GLTF Models, Browser-Based Hand Tracking

CERTIFICATIONS

One Million Prompters | Dubai Future Foundation | 2026

Introduction in Artificial Intelligence | Edraak | 2025

Ignition 8.3 Credential | Inductive University | 2026

TECHNICAL EXPERIENCE

Researcher, MS Lab - Robotics & Advanced Projects | University of Baghdad

Contributed to robotics prototypes, embedded integration, computer-vision workflows, interdisciplinary projects, and technical problem solving.

Technical Internships | Ministry of Electricity (2025) & Alchemy Scientific (2026), Baghdad

Gained exposure to SCADA/server systems, electrical control infrastructure, sensor calibration, testing, and basic troubleshooting of laboratory and water-quality instruments.

SELECTED PROJECTS

NeuroGuard 4.0 | AI + IIoT + SCADA + PLC | [View Project](#)

Developed YOLOv8 bottle-cap inspection with a custom ROI workflow, Flask JSON API, Node-RED logic, consecutive-defect detection, and SCADA/HMI alarms.

PetroSafe AI | Temporal Anomaly Detection | [View Project](#)

Built a tank-monitoring prototype using Factory I/O, Node-RED, Modbus TCP, FastAPI, and Isolation Forest for temporal anomaly scoring and fail-safe advisory decisions.

VisionSort AI | Computer Vision Automation | [View Project](#)

Integrated OpenCV, Python, Node-RED, Factory I/O, Modbus TCP, and HTTP for camera-based detection, automated sorting, and live monitoring.

Smart Ground Vehicle via BCI | Graduation Project | [View Project](#)

Integrated EEG signal processing, Python, ESP32, mobile control, an AI agent, and obstacle detection to translate mental commands into vehicle actions.

LANGUAGES

Arabic - Native | English - Intermediate